CHEM 1301 Exam 3 — Master Study Guide

Chapter 7 (Reactions in Aqueous Solutions) · Chapter 8 (Chemical Composition) · Chapter 9 (Chemical Quantities)

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How to use this guide

- **Exam format:** 30–35 multiple-choice questions, choices a–e. You get a periodic table. **You do not get the solubility rules, so memorize Table 7.1** (Section 7.3).
- **Mix:** expect about half concept questions (definitions, classifications, "which is true") and half calculations.
- Each section lists the slides it comes from (`Chapter N post.pdf` / `.pptx`, same slide numbers). **Definitions are in the course's wording.** When a textbook or website words something differently, go with the slides.
- Every "Worked example" is a practice problem from the slides, solved step by step. Cover the solution, try it, then check.
- After each practice exam, use its key's "Source" line to jump back to the matching section here.

Chapter 7 — Reactions in Aqueous Solutions

7.1 Driving forces (Ch 7 slides 2–4)

A reaction tends to happen when it forms one of these. Memorize all four.

1. **Formation of a solid** (precipitation)
2. **Formation of water** (acid–base)
3. **Transfer of electrons** (oxidation–reduction)
4. **Formation of a gas** (gas evolution)

7.2 Ions in water and electrolytes (Ch 7 slides 6–10)

- When an ionic solid dissolves, **the ions separate and move around independently**. $Ba(NO_3)_2(aq)$ contains separated Ba^{2+} and NO_3^- ions (two NO_3^- for every Ba^{2+}), not $Ba(NO_3)_2$ units.
- Pure water does **not** conduct electricity. Ions must be present for current to flow.
- **Strong electrolyte:** a substance where every unit that dissolves produces separated ions.
- **A solid compound must have zero net charge**, so a precipitate must contain both a cation and an anion. Two cations (K^+ and Ba^{2+}) or two anions (CrO_4^{2-} and NO_3^-) can never form a solid together (slide 10).

7.3 Solubility rules — MEMORIZE (Ch 7 slide 11, Table 7.1)

#	Rule	Exceptions to know
1	Most **nitrate** (NO_3^-) salts are **soluble**	—
2	Most salts of **Na^+, K^+, NH_4^+** are **soluble**	—
3	Most **chloride** salts are **soluble**	**AgCl, $PbCl_2$, Hg_2Cl_2** are insoluble
4	Most **sulfate** salts are **soluble**	**$BaSO_4$, $PbSO_4$, $CaSO_4$** are insoluble
5	Most **hydroxides** are only **slightly soluble**	**NaOH, KOH** soluble; $Ba(OH)_2$, $Ca(OH)_2$ moderately soluble
6	Most **sulfide** (S^{2-}), **carbonate** (CO_3^{2-}), **phosphate** (PO_4^{3-}) salts are only **slightly soluble**	—

"Insoluble" and "slightly soluble" mean the same thing here: too little dissolves to see.

Rules 1 and 2 beat rules 5 and 6. If a compound contains Na^+, K^+, NH_4^+, or NO_3^-, it is soluble, even when the other ion is a sulfide, carbonate, phosphate, or hydroxide. Examples: $(NH_4)_2S$, K_3PO_4, and Na_2CO_3 are all **soluble**.

Memory aid, in three tiers:

- **Always soluble:** NO_3^-, Na^+, K^+, NH_4^+.
- **Soluble except:** Cl^- (except **Ag, Pb, Hg_2**), SO_4^{2-} (except **Ba, Pb, Ca**). Pb is on both exception lists.
- **Insoluble except tier 1:** OH^- (NaOH, KOH soluble), S^{2-}, CO_3^{2-}, PO_4^{3-}.

Worked example (slide 12). Classify NaOH, $BaSO_4$, $CaCO_3$.

- NaOH: **soluble**. It is a named exception in rule 5, and rule 2 covers Na^+.
- $BaSO_4$: **insoluble**. It is a named exception in rule 4.
- $CaCO_3$: **insoluble**. Rule 6 covers carbonates, and Ca^{2+} is not a rule-2 cation.

HW14 style: "Which rule explains why ammonium sulfide is soluble?" Answer: *most salts of NH_4^+ are soluble* (rule 2), **not** the sulfide rule.

7.4 Predicting precipitates (Ch 7 slide 14)

1. **Write the reactants as they actually exist** (separated ions).
2. **Consider the solids that could form** by **exchanging the anions**.
3. **Use the solubility rules** to decide whether either product is insoluble. If neither is, write **NR** (no reaction).

Worked example (slides 15–17). $K_2CrO_4(aq) + Ba(NO_3)_2(aq)$. The ions are K^+, CrO_4^{2-}, Ba^{2+}, NO_3^-. Swapping anions gives KNO_3 (soluble, rules 1 and 2) and $BaCrO_4$, which is the solid on the slides.

$$K_2CrO_4(aq) + Ba(NO_3)_2(aq) \rightarrow BaCrO_4(s) + 2KNO_3(aq)$$

NR example (HW15). $KNO_3(aq) + Na_2S(aq)$. Swapping gives K_2S and $NaNO_3$. Both contain rule-2 cations, so both are soluble and the answer is **NR**.

7.5 Three kinds of equations (Ch 7 slides 15–20)

Equation	Shows	Spectator ions?
Molecular	complete formulas of all reactants and products; doesn't show the actual forms in solution	n/a
Complete ionic	every **strong electrolyte** written as separated ions; **all** reactants and products included	included
Net ionic	**only the components that undergo a change** (are directly involved)	**left out**

- **Spectator ions:** ions that don't participate directly in the reaction. They appear unchanged on both sides.
- Write as ions: soluble ionic compounds (aq), strong acids, strong bases. Keep as full formulas: solids (s), water (l), gases (g), and weak acids such as HNO_2.

Worked example (slide 19). $Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$

- Complete ionic: $Pb^{2+}(aq) + 2NO_3^-(aq) + 2K^+(aq) + 2I^-(aq) \rightarrow PbI_2(s) + 2K^+(aq) + 2NO_3^-(aq)$
- Cancel the spectators K^+ and NO_3^-.
- Net ionic: $\mathbf{Pb^{2+}(aq) + 2I^-(aq) \rightarrow PbI_2(s)}$

Worked example (slide 20). Sodium chloride solution added to silver nitrate solution.

- Molecular: $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$. AgCl is a rule-3 exception.
- Complete ionic: $Na^+(aq) + Cl^-(aq) + Ag^+(aq) + NO_3{}^-(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3{}^-(aq)$
- Net ionic: $\mathbf{Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)}$

Net ionic checklist: right states, the smallest whole-number coefficients, and both charge and atoms balanced.

7.6 Acids, bases, and acid–base reactions (Ch 7 slides 21–28)

Course definitions

- **Arrhenius acid:** a substance that produces $\mathbf{H^+}$ **ions** (protons) when dissolved in water.
- **Arrhenius base:** a substance that produces **hydroxide ions** ($\mathbf{OH^-}$) in water.
- **Strong acid:** a substance that **completely dissociates (ionizes)** in water. Each molecule breaks into H^+ plus an anion. Because strong acids are strong electrolytes that produce H^+, they are called strong acids.
- **Strong base:** a **metal hydroxide that is very soluble in water** and completely breaks up into separated ions. Strong bases are **strong electrolytes**.
- **Salt:** the ionic compound formed along with water in an acid–base reaction. It may stay dissolved or precipitate.

Memorize the lists:

- **Strong acids: HCl, $\mathbf{HNO_3}$, $\mathbf{H_2SO_4}$.** Anything else here (HNO_2, HF, $HC_2H_3O_2$, H_3PO_4) is not on the list.
- **Strong bases: NaOH, KOH.** (HW16 also uses CsOH, another very soluble group-1 hydroxide.)

The six principles (slides 24–26), condensed

1. The common strong acids are HCl, HNO_3, H_2SO_4.
2. A strong acid completely dissociates.
3. A strong base is a very soluble metal hydroxide (NaOH, KOH) that completely breaks into ions.
4. **Strong acid + strong base always has the same net ionic equation: $\mathbf{H^+(aq) + OH^-(aq) \rightarrow H_2O(l)}$.**
5. One product is always water. The other is a **salt** that stays dissolved and can be recovered by evaporating the water.
6. H^+ is the acidic ion and OH^- is the basic ion.

Naming the salt (HW16): the base's cation goes with the acid's anion. HI + NaOH gives **NaI**, NaOH + HNO_3 gives $\mathbf{NaNO_3}$, and HCl + CsOH gives **CsCl**.

Worked example (slide 27). Balance $H_3PO_4 + Ca(OH)_2 \rightarrow Ca_3(PO_4)_2 + H_2O$.

1. Balance Ca: 3 $Ca(OH)_2$.
2. Balance PO_4 as a unit: 2 H_3PO_4.
3. Balance H and O: 6 H^+ + 6 OH^- make 6 H_2O.

$$2H_3PO_4 + 3Ca(OH)_2 \rightarrow Ca_3(PO_4)_2(s) + 6H_2O(l)$$

$Ca_3(PO_4)_2$ is a phosphate (rule 6), so it precipitates. Phosphoric acid isn't on the strong-acid list, so leave it molecular in the ionic equations:

- Net ionic: $2H_3PO_4(aq) + 3Ca^{2+}(aq) + 6OH^-(aq) \rightarrow Ca_3(PO_4)_2(s) + 6H_2O(l)$

Worked example (slide 28). Nitrous acid + potassium hydroxide.

- Molecular: $HNO_2(aq) + KOH(aq) \rightarrow KNO_2(aq) + H_2O(l)$, which is already balanced.
- HNO_2 is not a strong acid, so it stays molecular. KOH and KNO_2 are written as ions.
- Complete ionic: $HNO_2(aq) + K^+(aq) + OH^-(aq) \rightarrow K^+(aq) + NO_2{}^-(aq) + H_2O(l)$
- Net ionic: $\mathbf{HNO_2(aq) + OH^-(aq) \rightarrow NO_2{}^-(aq) + H_2O(l)}$. Only a **strong** acid gives $H^+ + OH^- \rightarrow H_2O$.

7.7 Oxidation–reduction (Ch 7 slides 29–34, 38)

- **Oxidation–reduction (redox) reaction:** a reaction that involves a **transfer of electrons**. One species loses electrons and another gains them (HW16).
- **Principle 1: metal + nonmetal always makes an ionic compound and is always redox.** The metal loses electrons to become a cation, and the nonmetal gains them to become an anion.
- **Principle 2:** two nonmetals can also undergo redox. **For now, you recognize these by O_2 appearing as a reactant or product.** The compound formed by two nonmetals is not ionic.

How many electrons? Use the charges in the ionic product. The number of electrons lost per metal atom equals the cation's charge.

Reaction	Ions formed	Electrons
$2Li + F_2 \rightarrow 2LiF$ (slide 38)	Li^+, F^-	each Li loses 1; each F gains 1
$Ba + Cl_2 \rightarrow BaCl_2$ (HW16)	Ba^{2+}, Cl^-	Ba loses 2; Cl_2 gains 2 (1 per Cl)
$\mathbf{2Al + N_2 \rightarrow 2AlN}$ **(slide 33)**	Al^{3+}, N^{3-}	**each Al loses 3**; each N gains 3

Worked example (slide 34). Solid copper reacts with aqueous iron(II) chloride to form solid iron and aqueous copper(II) chloride.

$$Cu(s) + FeCl_2(aq) \rightarrow Fe(s) + CuCl_2(aq)$$

This is already balanced. Cu (0) becomes Cu^{2+}, so **Cu loses 2 e$^-$**. Fe^{2+} becomes Fe (0), so **Fe^{2+} gains 2 e$^-$**. Cl^- doesn't change; it is a spectator.

7.8 Classifying reactions (Ch 7 slides 35–45)

Class	How to spot it	Slide example
Precipitation	two solutions make a **solid**	$K_2CrO_4 + Ba(NO_3)_2 \rightarrow BaCrO_4(s) + 2KNO_3$
Double-displacement (double-exchange)	two anions **swap** partners	same as above
Acid–base	H^+ ends up in **water**	$HCl + KOH \rightarrow H_2O + KCl$

Class	How to spot it	Slide example
Oxidation–reduction	**electrons transferred** (metal + nonmetal, or O_2 involved)	$2Li + F_2 \rightarrow 2LiF$
Gas evolution	a **gas** forms (also acid–base here)	$2HCl + Na_2CO_3 \rightarrow CO_2(g) + H_2O + 2NaCl$
Single-replacement	**one** anion moves from one cation to another (always redox)	$Zn + 2HCl \rightarrow ZnCl_2 + H_2$
Combustion	reacts with **oxygen**, releasing energy so fast that a **flame** results	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
Synthesis (combination)	a compound forms from **simpler materials**	$2Na + Cl_2 \rightarrow 2NaCl$
Decomposition	a compound breaks into simpler compounds or elements, usually by **heat or electric current**	$2HgO \rightarrow 2Hg + O_2$

One reaction can belong to several classes (HW17):

- $C + O_2 \rightarrow CO_2$ is synthesis, combustion, and redox.
- $2Fe + 3Cl_2 \rightarrow 2FeCl_3$ is synthesis and redox, but **not** combustion (no O_2).
- Combustion and synthesis/decomposition with elements are all redox. Combustion also produces **heat/energy** (HW17).

Worked example (slide 44).

- $N_2 + 3H_2 \rightarrow 2NH_3$ is **synthesis**.
- $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ is **combustion**.
- $2H_2O \rightarrow 2H_2 + O_2$ is **decomposition**.

Balancing from words (HW17):

- Phosphorus + chlorine: $P_4 + 6Cl_2 \rightarrow 4PCl_3$
- Ammonium nitrite decomposes: $NH_4NO_2 \rightarrow N_2 + 2H_2O$
- Hydrobromic acid decomposes: $2HBr \rightarrow H_2 + Br_2$

Chapter 8 — Chemical Composition

8.1 Counting by weighing; atomic mass units (Ch 8 slides 2–6)

- Objects that differ slightly can be **counted by weighing** if you know their **average mass** (the jellybean idea).
- **Atomic mass unit (slide 5):** a tiny unit of mass made for weighing single atoms. 1 amu = 1.66×10^{-24} g.
- **Average atomic mass (Table 8.1):** the mass of **ONE atom** of the element, in **amu**. It is an *average* only because an element's atoms are a mix of isotopes with slightly different masses (like jellybeans of slightly different sizes), so the number is the mass of a typical single atom. Table 8.1: H 1.008 amu, C 12.01 amu, N 14.01 amu, O 16.00 amu, Na 22.99 amu, Al 26.98 amu.

Number from the periodic table (C: 12.01) read as	**Means**	**Unit**
Average atomic mass	mass of **1 carbon atom**	**amu**
Molar mass (section 8.2)	mass of **1 mole** (6.022×10^{23}) of carbon atoms	**g/mol**

Mass of one atom in grams: multiply its amu value by 1.66×10^{-24} g/amu.

$$1 \text{ C atom} = 12.01 \text{ amu} \times \frac{1.66 \times 10^{-24} \text{ g}}{1 \text{ amu}} = 1.99 \times 10^{-23} \text{ g}$$

Check that the two readings agree: 6.022×10^{23} atoms $\times$ 1.99×10^{-23} g per atom = 12.0 g, so one mole of carbon weighs 12.01 g. That's why the same number serves as amu per atom and as grams per mole.

- **Key idea (slide 6):** samples whose masses are in the **same ratio as their atomic masses** contain the **same number of atoms**. So 26.98 g Al and 63.55 g Cu contain the same number of atoms.

8.2 The mole (Ch 8 slides 7–9)

- **Mole (course definition):** the number equal to the number of carbon atoms in **12.01 g of carbon**.
- **Avogadro's number:** 6.022×10^{23}. One mole of anything is 6.022×10^{23} units of it.
- 1 mol of any element has a mass in grams equal to its atomic mass (Table 8.2): Al 26.98 g, Au 196.97 g, Fe 55.85 g, S 32.07 g, B 10.81 g, Xe 131.3 g, and every one of these contains 6.022×10^{23} atoms.

8.3 The conversion map (Ch 8 slides 10–15)

$$\text{grams} \xrightarrow{\div \text{ molar mass}} \text{moles} \xrightarrow{\times\, 6.022\times10^{23}} \text{atoms or molecules}$$

Go backward by doing the opposite operations. **Moles are always the middle step.** For mg, μg, or kg, convert to grams first.

The two conversion factors on this map:

Factor	Converts	Where the number comes from
$\dfrac{1 \text{ mol X}}{\text{atomic mass of X in g}}$ (or flipped)	grams ↔ moles	The element's atomic mass on the periodic table, read as **g/mol** (section 8.1)
$\dfrac{6.022 \times 10^{23} \text{ atoms}}{1 \text{ mol}}$ (or flipped)	moles ↔ atoms	Avogadro's number: 1 mol is always 6.022×10^{23} things (section 8.2)

Put the unit you **want** on top and the unit you **have** on the bottom so the old unit cancels.

Worked example (slide 10). How many H atoms are in 0.500 g of H?

Step 1: grams to moles. The atomic mass of H is 1.008, so 1 mol H weighs 1.008 g. You have grams, so grams goes on the bottom:

$$0.500 \text{ g H} \times \underbrace{\frac{1 \text{ mol H}}{1.008 \text{ g H}}}_{\text{atomic mass of H}} = 0.496 \text{ mol H}$$

Step 2: moles to atoms. 1 mol is 6.022×10^{23} atoms. You have moles, so moles goes on the bottom:

$$0.496 \text{ mol H} \times \underbrace{\frac{6.022 \times 10^{23} \text{ H atoms}}{1 \text{ mol H}}}_{\text{Avogadro's number}} = \mathbf{2.99 \times 10^{23} \text{ H atoms}}$$

Worked example (slide 11). 3.60 g Al to moles. The atomic mass of Al is 26.98, so 1 mol Al = 26.98 g:

$$3.60 \text{ g Al} \times \underbrace{\frac{1 \text{ mol Al}}{26.98 \text{ g Al}}}_{\text{atomic mass of Al}} = \mathbf{0.133 \text{ mol Al}}$$

Worked example (slide 12). 544 g Ca to moles. The atomic mass of Ca is 40.08:

$$544 \text{ g Ca} \times \underbrace{\frac{1 \text{ mol Ca}}{40.08 \text{ g Ca}}}_{\text{atomic mass of Ca}} = \mathbf{13.6 \text{ mol Ca}}$$

Worked example (slide 13). 0.133 mol Al (from slide 11) to atoms:

$$0.133 \text{ mol Al} \times \underbrace{\frac{6.022 \times 10^{23} \text{ Al atoms}}{1 \text{ mol Al}}}_{\text{Avogadro's number}} = \mathbf{8.01 \times 10^{22} \text{ Al atoms}}$$

Worked example (slide 14). 13.6 mol Ca (from slide 12) to atoms:

$$13.6 \text{ mol Ca} \times \underbrace{\frac{6.022 \times 10^{23} \text{ Ca atoms}}{1 \text{ mol Ca}}}_{\text{Avogadro's number}} = \mathbf{8.19 \times 10^{24} \text{ Ca atoms}}$$

Worked example (slide 15): going backward, atoms to grams. What is the mass of 2.31×10^{22} Au atoms? Atoms to moles uses Avogadro's number **flipped** (atoms on the bottom), then moles to grams uses the atomic mass of Au (196.97) **with grams on top**:

$$2.31 \times 10^{22} \cancel{\text{ Au atoms}} \times \underbrace{\frac{1 \text{ mol Au}}{6.022 \times 10^{23} \cancel{\text{ Au atoms}}}}_{\text{Avogadro's number, flipped}} = 0.0384 \text{ mol Au}$$

$$0.0384 \cancel{\text{ mol Au}} \times \underbrace{\frac{196.97 \text{ g Au}}{1 \cancel{\text{ mol Au}}}}_{\text{atomic mass of Au}} = \mathbf{7.56 \text{ g Au}}$$

Reality check (slide 19): a sample smaller than 1 molar mass must be less than 1 mol and fewer than 6.022×10^{23} atoms. An atom count around 10^{-20} or 10^{50} means you flipped a factor upside down.

HW18 style: "A sample of Co contains 3.10×10^{24} atoms; a sample of Mn contains 4.10 mol. Which has more atoms?" You can only compare amounts in the same unit, so convert the Mn to atoms:

$$4.10 \cancel{\text{ mol Mn}} \times \underbrace{\frac{6.022 \times 10^{23} \text{ Mn atoms}}{1 \cancel{\text{ mol Mn}}}}_{\text{Avogadro's number}} = 2.47 \times 10^{24} \text{ Mn atoms}$$

3.10×10^{24} Co atoms is more than 2.47×10^{24} Mn atoms, so the **Co** sample has more atoms.

8.4 Conceptual problem solving (Ch 8 slides 16–19)

1. **Where are we going?** Read the problem, identify the final goal, and sort the facts given (draw it if it helps).
2. **How do we get there?** Work backward from the goal. In stoichiometry, always start with the balanced equation.
3. **Reality check:** does the answer make sense?

8.5 Molar mass (Ch 8 slides 21–29)

- **Molar mass:** the mass in grams of **1 mole** of a substance, found by **summing the masses of the component atoms**.
- **A subscript outside parentheses multiplies everything inside.** $Ba(OH)_2$ has 1 Ba, 2 O, and 2 H.

How to calculate one: (1) count how many atoms of each element are in the formula (the subscript; no subscript means 1); (2) multiply each count by that element's atomic mass from the periodic table; (3) add the results.

Worked examples. Each term is (number of atoms) × (atomic mass of that element).

Slide	Compound	Atom count	(count × atomic mass), added	Molar mass
22	CH_4	1 C, 4 H	1(12.01) + 4(1.008)	**16.04 g/mol**
23	NH_3	1 N, 3 H	1(14.01) + 3(1.008)	**17.03 g/mol**

Slide	Compound	Atom count	(count × atomic mass), added	Molar mass
24	H_2SO_4	2 H, 1 S, 4 O	2(1.008) + 1(32.07) + 4(16.00)	**98.09 g/mol**
25	$C_6H_{12}O_6$	6 C, 12 H, 6 O	6(12.01) + 12(1.008) + 6(16.00)	**180.16 g/mol**
26	$Ba(OH)_2$	1 Ba, 2 O, 2 H	1(137.33) + 2(16.00) + 2(1.008)	**171.35 g/mol**

Mass and moles (slides 27–29). The molar mass is the conversion factor: 1 mol of the compound = (molar mass) g.

Slide 27. What is the mass of 7.89 mol NaOH? Molar mass of NaOH = 22.99 + 16.00 + 1.008 = 40.00 g/mol. You have moles and want grams, so grams goes on top:

$$7.89 \cancel{\text{ mol NaOH}} \times \underbrace{\frac{40.00 \text{ g NaOH}}{1 \cancel{\text{ mol NaOH}}}}_{\text{molar mass of NaOH}} = \mathbf{316 \text{ g NaOH}}$$

Slide 28. How many moles are in 982 g FeO? Molar mass of FeO = 55.85 + 16.00 = 71.85 g/mol. You have grams and want moles, so grams goes on the bottom:

$$982 \cancel{\text{ g FeO}} \times \underbrace{\frac{1 \text{ mol FeO}}{71.85 \cancel{\text{ g FeO}}}}_{\text{molar mass of FeO}} = \mathbf{13.7 \text{ mol FeO}}$$

Slide 29. How many moles are in 9.43 mg $CaCl_2$? Molar mass of $CaCl_2$ = 40.08 + 2(35.45) = 110.98 g/mol. The molar mass is in **grams**, so convert mg to g first (1 g = 1000 mg):

$$9.43 \cancel{\text{ mg}} \times \underbrace{\frac{1 \text{ g}}{1000 \cancel{\text{ mg}}}}_{\text{metric}} = 9.43 \times 10^{-3} \text{ g } CaCl_2$$

$$9.43 \times 10^{-3} \cancel{\text{ g } CaCl_2} \times \underbrace{\frac{1 \text{ mol } CaCl_2}{110.98 \cancel{\text{ g } CaCl_2}}}_{\text{molar mass of } CaCl_2} = \mathbf{8.50 \times 10^{-5} \text{ mol}}$$

HW19 style: name to formula to molar mass. Write the formula first, then add up its atoms. Iron(II) chloride = $FeCl_2$: 55.85 + 2(35.45) = **126.75 g/mol**. Copper(II) sulfate = $CuSO_4$: 63.55 + 32.07 + 4(16.00) = **159.62 g/mol**. Rubidium chloride = RbCl.

HW19 style: moles of an atom to molecules. "A sample of C_6H_6 contains 5.09 mol C. How many C_6H_6 molecules are there?" Each C_6H_6 molecule contains **6** C atoms (the subscript), so 1 mol C_6H_6 contains 6 mol C. Use that as the factor, then Avogadro's number:

$$5.09 \cancel{\text{ mol C}} \times \underbrace{\frac{1 \text{ mol } C_6H_6}{6 \cancel{\text{ mol C}}}}_{\text{subscript of C in } C_6H_6} = 0.848 \text{ mol } C_6H_6$$

$$0.848 \cancel{\text{ mol}} \times \underbrace{\frac{6.022 \times 10^{23} \text{ molecules}}{1 \cancel{\text{ mol}}}}_{\text{Avogadro's number}} = \mathbf{5.11 \times 10^{23} \text{ molecules}}$$

8.6 Mass percent (Ch 8 slides 30–33)

$$\text{mass percent of element} = \frac{\text{mass of the element in 1 mol of compound}}{\text{mass of 1 mol of compound}} \times 100\%$$

Method: (1) find the molar mass of the compound (section 8.5). (2) For each element, the top of the fraction is (number of that atom in the formula) × (its atomic mass). (3) Divide by the molar mass and multiply by 100.

Worked example (slide 31). Ethanol, C_2H_5OH. Count the atoms: 2 C, **6** H (5 + the 1 in OH), 1 O. Molar mass = 2(12.01) + 6(1.008) + 1(16.00) = 24.02 + 6.048 + 16.00 = 46.07 g/mol.

- C: $\frac{2(12.01) = 24.02 \text{ g C}}{46.07 \text{ g } C_2H_5OH} \times 100\% = \mathbf{52.14\%}$
- H: $\frac{6(1.008) = 6.048 \text{ g H}}{46.07 \text{ g } C_2H_5OH} \times 100\% = \mathbf{13.13\%}$
- O: $\frac{1(16.00) = 16.00 \text{ g O}}{46.07 \text{ g } C_2H_5OH} \times 100\% = \mathbf{34.73\%}$

These add to 100%, which checks the work.

Worked example (slide 32). H_3PO_4: 3 H, 1 P, 4 O. Molar mass = 3(1.008) + 30.97 + 4(16.00) = 97.99 g/mol.

- H: 3(1.008) = 3.024 g; 3.024 / 97.99 × 100 = **3.09%**
- P: 1(30.97) = 30.97 g; 30.97 / 97.99 × 100 = **31.60%**
- O: 4(16.00) = 64.00 g; 64.00 / 97.99 × 100 = **65.31%**

Worked example (slide 33). Magnesium iodite, $Mg(IO_2)_2$ (the iodite ion is IO_2^-). The subscript 2 outside the parentheses doubles everything inside, so there are 1 Mg, **2** I, and **4** O. Molar mass = 24.31 + 2(126.90) + 4(16.00) = 342.11 g/mol.

- Mg: 1(24.31) = 24.31 g; 24.31 / 342.11 × 100 = **7.11%**
- I: 2(126.90) = 253.80 g; 253.80 / 342.11 × 100 = **74.19%**
- O: 4(16.00) = 64.00 g; 64.00 / 342.11 × 100 = **18.71%**

Trap: count the atoms of that element in the formula. KNO_2 has 2 O, and PCl_5 has 5 Cl (HW20: P 14.87%, Cl 85.13%).

8.7 Empirical and molecular formulas (Ch 8 slides 34–50)

Definitions

- **Empirical (simplest) formula:** the **smallest whole-number ratio** of atoms in the compound.
- **Molecular formula:** the **actual** formula: how many of each atom are really in one molecule.
- **Molecular formula = (empirical formula)$_n$**, where n is a whole number (1, 2, 3, …). Glucose is $C_6H_{12}O_6$. Divide every subscript by 6 to get its empirical formula, CH_2O, so n = 6.
- **What data you need (slide 47, HW21):** the empirical formula needs the **composition by mass**. The molecular formula needs the composition **and the molar mass**.

There are **three different problem types** in this section. Figure out which one you have before you start.

You are given	You are asked for	Use
A molecular formula (e.g. $C_4H_8O_2$)	The empirical formula	**Type A**: reduce the subscripts. No masses needed.
Masses or mass percents of each element	The empirical formula	**Type B**: the 4-step mass → moles method
An empirical formula and a molar mass	The molecular formula	**Type C**: find n

Type A: molecular formula → empirical formula (slide 40)

You already have the atom counts, so **skip the mass and mole steps entirely**. Find the largest number that divides evenly into **every** subscript, and divide every subscript by it. A subscript you can't see is 1.

Molecular formula	Subscripts	Largest number dividing all of them	Empirical formula
H_2O_2	H 2, O 2	2	**HO**
$C_4H_8O_2$	C 4, H 8, O 2	2	$\mathbf{C_2H_4O}$
P_4O_{10}	P 4, O 10	2	$\mathbf{P_2O_5}$
$C_6H_{12}O_6$	C 6, H 12, O 6	6	$\mathbf{CH_2O}$
$C_7H_6O_2$ (benzoic acid, HW20)	C 7, H 6, O 2	1 (nothing divides 7, 6, and 2)	$\mathbf{C_7H_6O_2}$, unchanged

If the only number that divides every subscript is 1, the formula is **already empirical**.

Type B: masses or percents → empirical formula (slides 35–37, 42–46)

You only know **how many grams** of each element there are. A formula counts **atoms**, so first convert grams to moles, then compare the moles.

Step 1: Get grams of each element. If you're given percents, pretend you have **exactly 100 g** of the compound. Then each percent is that many grams: 32.0% C becomes 32.0 g C. If you're given grams, use them as they are.

Step 2: Convert each element's grams to moles. Do this separately for each element, using that element's atomic mass from the periodic table:

$$\text{mol of element} = \frac{\text{grams of that element}}{\text{atomic mass of that element (g/mol)}}$$

Step 3: Divide every mole value by the smallest mole value. Look at the moles from step 2 and find the **smallest** one. Then divide **each** element's moles, including the smallest itself, by that one number. The smallest always becomes exactly 1, and the others tell you how many of that atom there are per 1 of the smallest.

Step 4: Turn the step-3 results into whole-number subscripts.

- **"Whole number" means within about 0.1 of an integer.** Measured data and rounded atomic masses add a little error, so 1.997, 2.03, and 0.98 all count as whole: round them to 2, 2, and 1.
- If a result ends in one of the decimals below, **don't round it**. It's a real fraction, so multiply.
- **Pick ONE multiplier and multiply EVERY element's result by it**, including the ones that were already whole. Use the smallest multiplier that makes **all** of them whole at the same time. (Using a different multiplier for each element would change the ratio and give a wrong formula.)

Decimal part of a step-3 result	Meaning	Multiply every result by
about .5	halves	**2**
about .33 or .67	thirds	**3**
about .25 or .75	quarters	**4**
about .2, .4, .6, or .8	fifths	**5**

If two elements need different multipliers, use the smallest number that works for both. For example, a .5 and a .33 together need 6.

The final whole numbers are the subscripts.

Worked example A, all whole after step 3 (slide 38). C 9.94%, Cl 58.6%, F 31.4%.

Element	Step 1: g (100 g sample)	Step 2: mol = g ÷ atomic mass	Step 3: ÷ smallest (0.828)	Step 4
C	9.94	9.94 ÷ 12.01 = 0.828	0.828 ÷ 0.828 = 1.00	1
Cl	58.6	58.6 ÷ 35.45 = 1.653	1.653 ÷ 0.828 = 1.997	2
F	31.4	31.4 ÷ 19.00 = 1.653	1.653 ÷ 0.828 = 1.997	2

1.997 is within 0.1 of 2, so round it. No multiplying is needed. The empirical formula is $\mathbf{CCl_2F_2}$.

Worked example B, a .5 forces ×2. An iron oxide is 69.94% Fe and 30.06% O.

Element	Step 1: g	Step 2: mol	Step 3: ÷ smallest (1.252)	Step 4: × 2
Fe	69.94	69.94 ÷ 55.85 = 1.252	1.00	1.00 × 2 = **2**
O	30.06	30.06 ÷ 16.00 = 1.879	1.50	1.50 × 2 = **3**

1.50 is not within 0.1 of a whole number, and it ends in .5, so multiply **both** results by 2. The formula is $\mathbf{Fe_2O_3}$. Rounding 1.5 to 2 instead would give FeO_2, which is wrong.

Worked example C, a .33 forces ×3, applied to all three elements. Vitamin C is 40.92% C, 4.58% H, 54.50% O.

Element	Step 1: g	Step 2: mol	Step 3: ÷ smallest (3.406)	Step 4: × 3
C	40.92	40.92 ÷ 12.01 = 3.407	1.00	**3**
H	4.58	4.58 ÷ 1.008 = 4.544	1.334	4.00, so **4**
O	54.50	54.50 ÷ 16.00 = 3.406	1.00	**3**

Only H came out as a fraction, but C and O get multiplied by 3 too, because the ratio has to stay the same. The empirical formula is $\mathbf{C_3H_4O_3}$. Writing CH_4O, with only H multiplied, is the classic mistake.

Worked example D, starting from grams (slides 45–46). 1.198 g C combines with 0.302 g H.

Element	Step 1: g	Step 2: mol	Step 3: ÷ smallest (0.0998)	Step 4
C	1.198	1.198 ÷ 12.01 = 0.0998	1.00	1
H	0.302	0.302 ÷ 1.008 = 0.300	3.00	3

The empirical formula is $\mathbf{CH_3}$. You were given grams, so step 1 is just "use the given grams".

Worked example E, four elements (slides 35–37). 32.000% C, 6.714% H, 42.630% O, 18.660% N. The moles are C 2.664, H 6.648, O 2.664, N 1.333. The smallest is 1.333, and dividing each by it gives 2, 5, 2, 1, so the formula is $\mathbf{C_2H_5O_2N}$.

HW21 style: Mg 20.19%, S 26.64%, O 53.17% gives $\mathbf{MgSO_4}$. Si 10.85%, Cl 27.40%, Br 61.75% gives $\mathbf{SiCl_2Br_2}$. Work them with the same table to check yourself.

Combustion analysis (slide 44): a carbon-containing sample is burned in excess O_2. The masses of the products give the percent of C (and the other elements), and from there it's a Type B problem.

Type C: empirical formula + molar mass → molecular formula (slides 47–49)

1. Add up the **empirical formula mass** (the molar mass of the empirical formula).
2. Divide: $n = \dfrac{\text{molar mass of the compound}}{\text{empirical formula mass}}$. The answer will be very close to a whole number.
3. **Multiply every subscript in the empirical formula by n.**

Worked example (slide 49). Empirical formula $C_4H_5N_2O$, molar mass 194.14 g/mol.

1. Empirical mass = 4(12.01) + 5(1.008) + 2(14.01) + 16.00 = 97.10 g/mol.
2. n = 194.14 ÷ 97.10 = 2.00
3. $C_{4\times2}H_{5\times2}N_{2\times2}O_{1\times2}$ = $\mathbf{C_8H_{10}N_4O_2}$ (caffeine)

HW21 style:

- CH with molar mass about 26: CH = 13.02, n = 2, so $\mathbf{C_2H_2}$.
- C_3H_7NO with molar mass 145–150: empirical mass 73.10, n = 2, so $\mathbf{C_6H_{14}N_2O_2}$.
- Vitamin C from example C, $C_3H_4O_3$ (88.06) with molar mass 176.12: n = 2, so $\mathbf{C_6H_8O_6}$.

Why the molecular formula is more useful (slide 50): it gives the actual number of atoms in a molecule. The empirical formula only gives the ratio, and many different compounds share one empirical formula (C_2H_2 and C_6H_6 are both CH).

Empirical-formula mistakes to avoid

1. Using grams or percents directly as subscripts. **Always convert to moles first.**
2. Dividing by something other than the **smallest mole value**, such as the smallest mass or the molar mass.
3. Rounding 1.5 to 2 or 1.33 to 1. Only round results within about 0.1 of a whole number.
4. Multiplying only the fractional element. **The same multiplier goes on every element.**
5. In Type C, multiplying only some of the subscripts by n.

Chapter 9 — Chemical Quantities

9.0 How to read the worked examples in this chapter

Every Chapter 9 problem is a chain of **conversion factors**. A conversion factor is a fraction whose top and bottom are equal amounts, so multiplying by it changes the **unit** without changing the **amount**. Only two kinds appear:

Kind of factor	Where its numbers come from	Example
Molar mass (grams ↔ moles)	Add up the atomic masses on the periodic table (section 8.5)	$\frac{58.44\text{ g NaCl}}{1\text{ mol NaCl}}$
Mole ratio (moles of one substance ↔ moles of another)	The **coefficients** of those two substances in the **balanced** equation	$\frac{2\text{ mol NaCl}}{1\text{ mol Cl}_2}$

Orientation rule: write the factor with the unit you **want** on top and the unit you **have** on the bottom. The unit on the bottom cancels the unit you're carrying, and you're left with the unit on top. In the examples below, the labels under each factor say where its numbers came from, and crossed-out units are the ones that cancel.

9.1 What a balanced equation tells you (Ch 9 slides 2–5)

- A balanced equation gives the **identities** of the reactants and products and **how much** of each takes part.
- **Coefficients give relative numbers of molecules (or moles), never grams.** What matters is the *ratio* of the coefficients, not any single coefficient.
- $CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$ can be read as 1 molecule + 2 molecules → 1 molecule, as 1 dozen + 2 dozen → 1 dozen, or as **1 mol + 2 mol → 1 mol** (Table 9.1).

Worked example (slide 4). $2Al_2O_3 \rightarrow 4Al + 3O_2$. The coefficients are Al_2O_3 **2**, Al **4**, O_2 **3**.

(a) How many O_2 molecules come from 4 molecules of Al_2O_3? You have Al_2O_3 and want O_2, so the factor is (O_2 coefficient) over (Al_2O_3 coefficient):

$$4\ \cancel{Al_2O_3} \times \underbrace{\frac{3\ O_2}{2\ \cancel{Al_2O_3}}}_{\text{coefficients of } O_2 \text{ and } Al_2O_3} = \mathbf{6\ O_2\ molecules}$$

(b) If 8 atoms of Al are produced, how many O_2 molecules form? You have Al and want O_2:

$$8\ \cancel{Al} \times \underbrace{\frac{3\ O_2}{4\ \cancel{Al}}}_{\text{coefficients of } O_2 \text{ and Al}} = \mathbf{6\ O_2\ molecules}$$

9.2 Mole ratios (Ch 9 slides 5–9)

A **mole ratio** comes from the coefficients of **two** substances in a balanced equation. Every pair of substances gives one, and each one can be written right side up or upside down.

How to build one. From $2H_2O \rightarrow 2H_2 + O_2$ (coefficients: H_2O **2**, H_2 **2**, O_2 **1**):

Pair	Statement	The two conversion factors it gives
H_2O and H_2	2 mol H_2O makes 2 mol H_2	$\frac{2\text{ mol }H_2}{2\text{ mol }H_2O}$ or $\frac{2\text{ mol }H_2O}{2\text{ mol }H_2}$
H_2O and O_2	2 mol H_2O makes 1 mol O_2	$\frac{1\text{ mol }O_2}{2\text{ mol }H_2O}$ or $\frac{2\text{ mol }H_2O}{1\text{ mol }O_2}$
H_2 and O_2	2 mol H_2 forms with 1 mol O_2	$\frac{1\text{ mol }O_2}{2\text{ mol }H_2}$ or $\frac{2\text{ mol }H_2}{1\text{ mol }O_2}$

Which one to use: the one with the substance you **want** on top and the substance you **have** on the bottom. For $4B + 3O_2 \rightarrow 2B_2O_3$, going from mol B to mol O_2 uses $\frac{3\text{ mol }O_2}{4\text{ mol B}}$ (HW22). The 3 is O_2's coefficient and the 4 is B's coefficient.

Worked example (slide 7). $N_2 + 3H_2 \rightarrow 2NH_3$ (coefficients N_2 **1**, H_2 **3**, NH_3 **2**). The mole ratios are 1 mol N_2 : 3 mol H_2, 1 mol N_2 : 2 mol NH_3, and 3 mol H_2 : 2 mol NH_3.

Worked example (slide 8). $N_2 + O_2 \rightarrow 2NO$ (coefficients N_2 **1**, O_2 **1**, NO **2**). The mole ratios are 1 mol N_2 : 1 mol O_2, 1 mol N_2 : 2 mol NO, and 1 mol O_2 : 2 mol NO.

Worked example (slide 9): balance first. Ethane burns: $C_2H_6 + O_2 \rightarrow CO_2 + H_2O$.

1. **C:** 2 C on the left, so put 2 in front of CO_2.
2. **H:** 6 H on the left, so put 3 in front of H_2O ($3 \times 2 = 6$ H).
3. **O:** the right now has $2(2) + 3(1) = 7$ O, so O_2 needs 7/2.
4. 7/2 isn't a whole number, so double **every** coefficient: $\mathbf{2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O}$.

9.3 Mole–mole conversions (Ch 9 slides 10–12)

One step: multiply the moles you have by the mole ratio (want over have).

$$\text{mol wanted} = \text{mol given} \times \frac{\text{coefficient of the substance wanted}}{\text{coefficient of the substance given}}$$

Worked example (slide 10). $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$. How many mol Fe form from 17.5 mol CO? Have CO (coefficient **3**); want Fe (coefficient **2**).

$$17.5\ \cancel{\text{mol CO}} \times \underbrace{\frac{2\text{ mol Fe}}{3\ \cancel{\text{mol CO}}}}_{\text{coefficients of Fe and CO}} = \mathbf{11.7\text{ mol Fe}}$$

Worked example (slide 11). $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$. How many mol O_2 react with 0.987 mol C_2H_6? Have C_2H_6 (coefficient **2**); want O_2 (coefficient **7**).

$$0.987 \cancel{\text{ mol } C_2H_6} \times \underbrace{\frac{7 \text{ mol } O_2}{2 \cancel{\text{ mol } C_2H_6}}}_{\text{coefficients of } O_2 \text{ and } C_2H_6} = \mathbf{3.45 \text{ mol } O_2}$$

Worked example (slide 12). $NH_3 + NaOCl \rightarrow NaOH + NH_2Cl$. How much NH_2Cl forms from 1.32 mol NH_3? Both coefficients are **1** (no number written means 1), so the ratio is 1 : 1.

$$1.32 \cancel{\text{ mol } NH_3} \times \underbrace{\frac{1 \text{ mol } NH_2Cl}{1 \cancel{\text{ mol } NH_3}}}_{\text{coefficients}} = 1.32 \text{ mol } NH_2Cl$$

To get grams, multiply by the molar mass of $NH_2Cl = 14.01 + 2(1.008) + 35.45 = 51.48$ g/mol:

$$1.32 \cancel{\text{ mol } NH_2Cl} \times \underbrace{\frac{51.48 \text{ g } NH_2Cl}{1 \cancel{\text{ mol } NH_2Cl}}}_{\text{molar mass of } NH_2Cl} = \mathbf{67.9 \text{ g } NH_2Cl}$$

9.4 Stoichiometry with masses (Ch 9 slides 13–20)

Stoichiometry: using a balanced chemical equation to calculate the relative masses of reactants and products. **The equation must be balanced first**, because the mole ratios come from the coefficients (slide 16).

Steps (slide 14)

1. Balance the equation.
2. Convert the grams you're given to **moles** (divide by that substance's molar mass).
3. Set up the **mole ratio** (want over have, from the coefficients).
4. Multiply to get the **moles** of the substance you want.
5. Convert those moles to **grams** (multiply by the wanted substance's molar mass).

$$\text{g A} \xrightarrow{\times \frac{1 \text{ mol A}}{\text{MM of A}}} \text{mol A} \xrightarrow{\times \frac{\text{coef B}}{\text{coef A}}} \text{mol B} \xrightarrow{\times \frac{\text{MM of B}}{1 \text{ mol B}}} \text{g B}$$

You can never go straight from grams of A to grams of B. The coefficients count moles, so you have to pass through moles.

Worked example (slide 17). $2SO_2 + O_2 \rightarrow 2SO_3$. What mass of SO_3 forms from 2.33 mol O_2?

- **Given:** 2.33 mol O_2. It's already in moles, so skip step 2.
- **Mole ratio:** have O_2 (coefficient **1**), want SO_3 (coefficient **2**).
- **Molar mass of SO_3:** $32.07 + 3(16.00) = 80.07$ g/mol.

$$2.33 \cancel{\text{ mol } O_2} \times \underbrace{\frac{2 \text{ mol } SO_3}{1 \cancel{\text{ mol } O_2}}}_{\text{coefficients}} = 4.66 \text{ mol } SO_3$$

$$4.66 \cancel{\text{ mol } SO_3} \times \underbrace{\frac{80.07 \text{ g } SO_3}{1 \cancel{\text{ mol } SO_3}}}_{\text{molar mass of } SO_3} = \mathbf{373 \text{ g } SO_3}$$

Worked example (slide 18). $H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$. How many moles of H_2SO_4 react with 19.9 g KOH?

- **Molar mass of KOH:** 39.10 + 16.00 + 1.008 = 56.11 g/mol.
- **Mole ratio:** have KOH (coefficient **2**), want H_2SO_4 (coefficient **1**).
- The question asks for **moles**, so stop after the mole ratio (no step 5).

$$19.9 \text{ g KOH} \times \underbrace{\frac{1 \text{ mol KOH}}{56.11 \text{ g KOH}}}_{\text{molar mass of KOH}} = 0.3547 \text{ mol KOH}$$

$$0.3547 \text{ mol KOH} \times \underbrace{\frac{1 \text{ mol } H_2SO_4}{2 \text{ mol KOH}}}_{\text{coefficients}} = \mathbf{0.177 \text{ mol } H_2SO_4}$$

Worked example (slide 19). $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$. What mass of NH_3 is needed to make 54.7 g NO?

- **Molar masses:** NO = 14.01 + 16.00 = 30.01 g/mol; NH_3 = 14.01 + 3(1.008) = 17.03 g/mol.
- **Mole ratio:** have NO (coefficient **4**), want NH_3 (coefficient **4**). It's 4 : 4, so the moles are equal.

$$54.7 \text{ g NO} \times \underbrace{\frac{1 \text{ mol NO}}{30.01 \text{ g NO}}}_{\text{molar mass of NO}} = 1.823 \text{ mol NO}$$

$$1.823 \text{ mol NO} \times \underbrace{\frac{4 \text{ mol } NH_3}{4 \text{ mol NO}}}_{\text{coefficients}} \times \underbrace{\frac{17.03 \text{ g } NH_3}{1 \text{ mol } NH_3}}_{\text{molar mass of } NH_3} = \mathbf{31.0 \text{ g } NH_3}$$

Worked example (slide 20). $CuSO_4 + Fe \rightarrow FeSO_4 + Cu$. What mass of $CuSO_4$ reacts with 244 g Fe?

- **Molar masses:** Fe = 55.85 g/mol; $CuSO_4$ = 63.55 + 32.07 + 4(16.00) = 159.62 g/mol.
- **Mole ratio:** have Fe (coefficient **1**), want $CuSO_4$ (coefficient **1**).

$$244 \text{ g Fe} \times \underbrace{\frac{1 \text{ mol Fe}}{55.85 \text{ g Fe}}}_{\text{molar mass of Fe}} = 4.369 \text{ mol Fe}$$

$$4.369 \text{ mol Fe} \times \underbrace{\frac{1 \text{ mol } CuSO_4}{1 \text{ mol Fe}}}_{\text{coefficients}} \times \underbrace{\frac{159.62 \text{ g } CuSO_4}{1 \text{ mol } CuSO_4}}_{\text{molar mass of } CuSO_4} = \mathbf{697 \text{ g } CuSO_4}$$

HW23 style (balance first, work in mg). $Fe + Cl_2 \rightarrow FeCl_3$. What mass of $FeCl_3$ forms from 20.5 mg Fe?

1. **Balance:** $FeCl_3$ has 3 Cl, and Cl_2 comes in pairs, so use $2FeCl_3$ (6 Cl) and $3Cl_2$ (6 Cl). Then 2 Fe on the left: $\mathbf{2Fe + 3Cl_2 \rightarrow 2FeCl_3}$.
2. **Molar masses:** Fe = 55.85; $FeCl_3$ = 55.85 + 3(35.45) = 162.20. A molar mass in g/mol is the same number in **mg/mmol** (both top and bottom are divided by 1000), so you can stay in mg and mmol.
3. **Mole ratio:** have Fe (coefficient **2**), want $FeCl_3$ (coefficient **2**).

$$20.5\ \cancel{\text{mg Fe}} \times \underbrace{\frac{1\ \text{mmol Fe}}{55.85\ \cancel{\text{mg Fe}}}}_{\text{molar mass of Fe}} \times \underbrace{\frac{2\ \text{mmol FeCl}_3}{2\ \cancel{\text{mmol Fe}}}}_{\text{coefficients}} \times \underbrace{\frac{162.20\ \text{mg FeCl}_3}{1\ \cancel{\text{mmol FeCl}_3}}}_{\text{molar mass of FeCl}_3} = \mathbf{59.5\ mg\ FeCl_3}$$

9.5 Limiting reactant (Ch 9 slides 21–24)

- **Limiting reactant (limiting reagent):** the reactant that is **used up first**. It **controls how much product forms**.
- The other reactant is **in excess**: some of it is left over.

Method

1. Convert every reactant to **moles**.
2. Pick one reactant. Use the mole ratio to find **how many moles of the other reactant it needs** to react completely.
3. Compare that **need** with what you **have** of the other reactant:
   - have **more** than needed: the other reactant is in excess, so the one you picked is **limiting**;
   - have **less** than needed: the other reactant runs out first, so **it** is limiting.

Cross-check (optional): calculate how many moles of product each reactant could make on its own. The reactant that gives the **smaller** amount is limiting.

Never compare grams, and never compare raw mole amounts without the mole ratio. A reactant with more moles can still be limiting if the equation needs a lot of it.

Worked example (slide 23). $3Mg + N_2 \rightarrow Mg_3N_2$, with 4.75 mol Mg and 2.11 mol N_2. Which is limiting?

Step 1: both are already in moles.

Step 2: pick Mg and find how much N_2 it needs. Have Mg (coefficient **3**), want N_2 (coefficient **1**):

$$4.75\ \cancel{\text{mol Mg}} \times \underbrace{\frac{1\ \text{mol N}_2}{3\ \cancel{\text{mol Mg}}}}_{\text{coefficients of N}_2\text{ and Mg}} = 1.58\ \text{mol N}_2\ \text{needed}$$

Step 3: you **have** 2.11 mol N_2 but only **need** 1.58 mol. There's more N_2 than the Mg can use, so N_2 is in excess and **Mg is limiting**. (Note that Mg has *more* moles, 4.75 vs 2.11, and is still limiting, because the equation uses 3 Mg for every 1 N_2.)

Worked example (slide 24). $4Al + 3O_2 \rightarrow 2Al_2O_3$, with 17.9 g Al and 64 g O_2. Which is limiting?

Step 1: grams to moles. Molar masses: Al = 26.98 g/mol; $O_2 = 2(16.00) = 32.00$ g/mol (O_2 has two O atoms).

$$17.9\ \cancel{\text{g Al}} \times \underbrace{\frac{1\ \text{mol Al}}{26.98\ \cancel{\text{g Al}}}}_{\text{molar mass of Al}} = 0.6635\ \text{mol Al}$$

$$64\ \cancel{\text{g O}_2} \times \underbrace{\frac{1\ \text{mol O}_2}{32.00\ \cancel{\text{g O}_2}}}_{\text{molar mass of O}_2} = 2.0\ \text{mol O}_2$$

Step 2: pick Al and find how much O_2 it needs. Have Al (coefficient **4**), want O_2 (coefficient **3**):

$$0.6635 \ \cancel{\text{mol Al}} \times \underbrace{\frac{3 \text{ mol } O_2}{4 \ \cancel{\text{mol Al}}}}_{\text{coefficients of } O_2 \text{ and Al}} = 0.498 \text{ mol } O_2 \text{ needed}$$

Step 3: you **have** 2.0 mol O_2 and only **need** 0.498 mol, so O_2 is in excess and **Al is limiting**. Al also has the smaller mass here, but that is *not* the reason. The comparison has to be made in moles through the mole ratio.

9.6 Theoretical yield (Ch 9 slides 25–27)

Theoretical yield: the amount of product **calculated from the limiting reactant**. It is the **maximum** amount of product that can be made from the amounts of reactants used.

Method

1. Find the **limiting reactant** (section 9.5). If the problem says one reactant is "in excess," the **other** one is limiting.
2. Start from the **moles of the limiting reactant**. Never start from the excess reactant.
3. Multiply by the mole ratio $\dfrac{\text{coefficient of the product}}{\text{coefficient of the limiting reactant}}$ to get moles of product.
4. Multiply by the **molar mass of the product** to get grams of product. That number is the theoretical yield.

Worked example (slide 26): one reactant in excess. $Pb(NO_3)_2 + 2NaCl \rightarrow PbCl_2 + 2NaNO_3$, with 237 g NaCl and excess $Pb(NO_3)_2$. What is the theoretical yield of $PbCl_2$?

Step 1: limiting reactant. $Pb(NO_3)_2$ is "in excess," which means there's more than enough of it. So **NaCl is limiting**, and no comparison is needed.

Step 2: grams NaCl to moles NaCl. Molar mass of NaCl = 22.99 + 35.45 = 58.44 g/mol.

$$237 \ \cancel{\text{g NaCl}} \times \underbrace{\frac{1 \text{ mol NaCl}}{58.44 \ \cancel{\text{g NaCl}}}}_{\text{molar mass of NaCl}} = 4.055 \text{ mol NaCl}$$

Step 3: moles NaCl to moles $PbCl_2$. In the balanced equation, NaCl's coefficient is **2** and $PbCl_2$'s coefficient is **1**. So 2 mol NaCl makes 1 mol $PbCl_2$. Want $PbCl_2$ on top, have NaCl on the bottom:

$$4.055 \ \cancel{\text{mol NaCl}} \times \underbrace{\frac{1 \text{ mol } PbCl_2}{2 \ \cancel{\text{mol NaCl}}}}_{\text{coefficient of } PbCl_2 \ / \ \text{coefficient of NaCl}} = 2.028 \text{ mol } PbCl_2$$

Step 4: moles $PbCl_2$ to grams $PbCl_2$. Molar mass of $PbCl_2$ = 207.2 + 2(35.45) = 278.1 g/mol.

$$2.028 \ \cancel{\text{mol PbCl}_2} \times \underbrace{\frac{278.1 \text{ g } PbCl_2}{1 \ \cancel{\text{mol PbCl}_2}}}_{\text{molar mass of } PbCl_2} = \mathbf{564 \text{ g } PbCl_2} \text{ (theoretical yield)}$$

Worked example (slide 27): you have to find the limiting reactant first. $2Na + Cl_2 \rightarrow 2NaCl$, with 68 g Na and 97 g Cl_2. What is the theoretical yield of NaCl?

Read the equation first. Coefficients: Na **2**, Cl_2 **1**, NaCl **2**. In words: 2 mol Na reacts with 1 mol Cl_2 to make 2 mol NaCl. The three mole ratios you can use are:

$$\frac{2 \text{ mol Na}}{1 \text{ mol } Cl_2} \qquad \frac{2 \text{ mol NaCl}}{2 \text{ mol Na}} \qquad \frac{2 \text{ mol NaCl}}{1 \text{ mol } Cl_2}$$

Step 1a: grams to moles for each reactant. Molar masses: Na = 22.99 g/mol (periodic table); Cl_2 = 2(35.45) = 70.90 g/mol (Cl_2 is diatomic, so it has **two** Cl atoms).

$$68 \text{ g Na} \times \underbrace{\frac{1 \text{ mol Na}}{22.99 \text{ g Na}}}_{\text{molar mass of Na}} = 2.958 \text{ mol Na}$$

$$97 \text{ g } Cl_2 \times \underbrace{\frac{1 \text{ mol } Cl_2}{70.90 \text{ g } Cl_2}}_{\text{molar mass of } Cl_2} = 1.368 \text{ mol } Cl_2$$

Step 1b: how much Na does all the Cl_2 need? Have Cl_2 (coefficient **1**), want Na (coefficient **2**):

$$1.368 \text{ mol } Cl_2 \times \underbrace{\frac{2 \text{ mol Na}}{1 \text{ mol } Cl_2}}_{\text{coefficient of Na / coefficient of } Cl_2} = 2.736 \text{ mol Na needed}$$

Step 1c: compare. You **have** 2.958 mol Na and the Cl_2 only **needs** 2.736 mol Na. There's extra Na (2.958 − 2.736 = 0.222 mol left over), so Na is in excess and the Cl_2 runs out first: **Cl_2 is limiting**.

Step 2: start from the limiting reactant, 1.368 mol Cl_2.

Step 3: moles Cl_2 to moles NaCl. Have Cl_2 (coefficient **1**), want NaCl (coefficient **2**). Here the 2 is **NaCl's** coefficient, not Na's. They happen to be the same number in this equation.

$$1.368 \text{ mol } Cl_2 \times \underbrace{\frac{2 \text{ mol NaCl}}{1 \text{ mol } Cl_2}}_{\text{coefficient of NaCl / coefficient of } Cl_2} = 2.736 \text{ mol NaCl}$$

Step 4: moles NaCl to grams NaCl. Molar mass of NaCl = 22.99 (Na) + 35.45 (Cl) = 58.44 g/mol.

$$2.736 \text{ mol NaCl} \times \underbrace{\frac{58.44 \text{ g NaCl}}{1 \text{ mol NaCl}}}_{\text{molar mass of NaCl}} = \textbf{160 g NaCl} \text{ (theoretical yield)}$$

All of steps 2–4 as one chain:

$$97 \text{ g } Cl_2 \times \frac{1 \text{ mol } Cl_2}{70.90 \text{ g } Cl_2} \times \frac{2 \text{ mol NaCl}}{1 \text{ mol } Cl_2} \times \frac{58.44 \text{ g NaCl}}{1 \text{ mol NaCl}} = 160 \text{ g NaCl}$$

Check with the cross-check method. If all the Na reacted, it would make 2.958 mol Na × (2 mol NaCl / 2 mol Na) × 58.44 g/mol = 173 g NaCl. That's more than the 160 g the Cl_2 allows, and the **smaller** amount wins. Using the Na would give 173 g, a wrong answer that is likely to appear among the choices, because there isn't enough Cl_2 to make that much.

9.7 Actual and percent yield (Ch 9 slides 28–32)

- **Actual yield:** the amount of product **actually produced**. It is usually less than the theoretical yield, for example because of **side reactions** that consume reactants or products.
- **Percent yield** shows how efficient a reaction is:

$$\text{percent yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

- The **actual yield is given** in the problem (it was measured). The **theoretical yield you calculate** (section 9.6). Both must be in the same unit, usually grams.
- **Can the actual yield be higher than the theoretical yield (slide 29)?** No. The theoretical yield is the maximum. A result over 100% means an error, such as a wet or impure product or a math mistake.

Worked example (slide 31). $H_2 + I_2 \rightarrow 2HI$, with 69.1 g H_2 and excess I_2. 134 g HI is actually produced. What is the percent yield?

Part 1: theoretical yield of HI. I_2 is in excess, so **H_2 is limiting**. Molar masses: $H_2 = 2(1.008) = 2.016$ g/mol; HI $= 1.008 + 126.90 = 127.91$ g/mol. Mole ratio: have H_2 (coefficient **1**), want HI (coefficient **2**).

$$69.1 \text{ g } \cancel{H_2} \times \underbrace{\frac{1 \text{ mol } H_2}{2.016 \text{ g } \cancel{H_2}}}_{\text{molar mass of } H_2} = 34.28 \text{ mol } H_2$$

$$34.28 \text{ } \cancel{\text{mol } H_2} \times \underbrace{\frac{2 \text{ mol HI}}{1 \text{ } \cancel{\text{mol } H_2}}}_{\text{coefficients}} \times \underbrace{\frac{127.91 \text{ g HI}}{1 \text{ } \cancel{\text{mol HI}}}}_{\text{molar mass of HI}} = 8768 \text{ g HI (theoretical)}$$

Part 2: percent yield. The actual yield (134 g) goes on top and the theoretical yield (8768 g) on the bottom:

$$\frac{134 \text{ g}}{8768 \text{ g}} \times 100\% = \mathbf{1.53}\%$$

Worked example (slide 32). $2KClO_3 \rightarrow 2KCl + 3O_2$, with 191 g $KClO_3$. 89.9 g KCl is actually produced. What is the percent yield?

Part 1: theoretical yield of KCl. $KClO_3$ is the only reactant, so it's limiting. Molar masses: $KClO_3 = 39.10 + 35.45 + 3(16.00) = 122.55$ g/mol; KCl $= 39.10 + 35.45 = 74.55$ g/mol. Mole ratio: have $KClO_3$ (coefficient **2**), want KCl (coefficient **2**).

$$191 \text{ g } \cancel{KClO_3} \times \underbrace{\frac{1 \text{ mol } KClO_3}{122.55 \text{ g } \cancel{KClO_3}}}_{\text{molar mass of } KClO_3} = 1.559 \text{ mol } KClO_3$$

$$1.559 \text{ } \cancel{\text{mol } KClO_3} \times \underbrace{\frac{2 \text{ mol KCl}}{2 \text{ } \cancel{\text{mol } KClO_3}}}_{\text{coefficients}} \times \underbrace{\frac{74.55 \text{ g KCl}}{1 \text{ } \cancel{\text{mol KCl}}}}_{\text{molar mass of KCl}} = 116.2 \text{ g KCl (theoretical)}$$

Part 2: percent yield.

$$\frac{89.9\text{ g}}{116.2\text{ g}} \times 100\% = \mathbf{77.4}\%$$

HW24 style (two-part). Some problems give amounts of **both** reactants (for example Xe + $2F_2 \rightarrow XeF_4$). Then part 1 needs the limiting reactant first (section 9.5), followed by the theoretical yield from that reactant (section 9.6). Part 2 is percent yield = actual / theoretical × 100. "Percentage of capacity reached" (the LiOH and CO_2 canister) uses the same formula: actual / theoretical × 100.

Common traps — check these before you bubble

1. **Solubility exceptions:** AgCl, $PbCl_2$, Hg_2Cl_2 and $BaSO_4$, $PbSO_4$, $CaSO_4$ are insoluble. $Na^+/K^+/NH_4{}^+/NO_3{}^-$ always win.
2. **Net ionic means no spectators.** A choice that still has Na^+ and $NO_3{}^-$ on both sides is the complete ionic equation.
3. **Precipitates are (s).** The same formula labeled (aq) is a wrong choice.
4. **Strong acids are HCl, HNO_3, H_2SO_4 only.** $HNO_2 \neq HNO_3$.
5. **Electrons per atom = ion charge**, not the total for the equation.
6. **More than one class can apply.** A single replacement is also redox. Burning in O_2 is combustion even when one product forms.
7. **mg $\rightarrow$ g first.** 1 mg = 10^{-3} g and 1 μg = 10^{-6} g.
8. **Subscripts outside parentheses multiply everything inside**: $Al_2(SO_4)_3$ has 12 O.
9. **Mass percent uses mass, not atom counts.** 1 P out of 8 atoms is not 12.5% P.
10. **Empirical formula:** divide by the *smallest* number of moles; multiply to clear 0.5, 0.33, or 0.25 fractions.
11. **Coefficients are moles, not grams.** Always convert to moles before using a ratio.
12. **Limiting reactant:** compare moles needed with moles available. Never compare grams or coefficients alone.
13. **Percent yield $\leq$ 100%.** Divide actual by **theoretical** yield, not by the mass of the reactant.

Glossary of course definitions

Term	Course definition (slide)
Precipitation / precipitate	Formation of a solid when two solutions are mixed; the solid is the precipitate (7: 5, 35)
Strong electrolyte	Each unit that dissolves produces separated ions (7: 7)
Molecular equation	Shows complete formulas of all reactants and products (7: 15)
Complete ionic equation	All strong electrolytes shown as ions; all reactants and products included (7: 16, 18)
Net ionic equation	Only the components directly involved / that undergo a change; no spectators (7: 17)
Spectator ions	Ions that don't participate directly in the reaction (7: 17)
Arrhenius acid / base	Produces H^+ / OH^- in water (7: 21, 22)
Strong acid	Completely dissociates (ionizes) in water: HCl, HNO_3, H_2SO_4 (7: 24)
Strong base	Metal hydroxide that is very soluble and completely breaks into ions: NaOH, KOH (7: 25)
Salt	Ionic compound formed with water in an acid–base reaction (7: 23, 26)
Oxidation–reduction	Reaction involving a transfer of electrons (7: 30)
Double-displacement	Two anions exchanged between cations (7: 36)

Term	Course definition (slide)
Single-replacement	One anion moves from one cation to another (7: 40)
Combustion	Reaction with oxygen that releases energy fast enough to make a flame (7: 41)
Synthesis (combination)	A compound forms from simpler materials (7: 42)
Decomposition	A compound breaks into simpler compounds or elements, by heat or current (7: 43)
Atomic mass unit	1 amu = 1.66×10^{-24} g (8: 5)
Mole	Number of C atoms in 12.01 g of C = 6.022×10^{23} (Avogadro's number) (8: 7)
Molar mass	Mass in grams of 1 mole of a substance; sum of the atomic masses (8: 22)
Mass percent	(Mass of element in 1 mol) / (mass of 1 mol) × 100% (8: 30)
Empirical formula	Smallest whole-number ratio of atoms (8: 39)
Molecular formula	Actual formula = $(\text{empirical})_{\text{n}}$ (8: 39, 48)
Stoichiometry	Using a balanced equation to calculate masses of reactants and products (9: 15)
Limiting reactant	The reactant used up first; it controls the amount of product (9: 21)
Theoretical yield	Product amount calculated from the limiting reactant; the maximum possible (9: 25, 28)
Actual yield	Product amount actually obtained (9: 28)
Percent yield	Actual / theoretical × 100% (9: 30)